

Experiment 20:

Implement Future Sales Prediction using a suitable machine learning algorithm

Python Code:

```
import numpy as np

from sklearn.linear_model import LinearRegression

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]])
y = np.array([120, 135, 150, 165, 180, 195, 210, 225, 240, 255, 270, 285])

model = LinearRegression()

model.fit(X, y)

future_months = np.array([[13], [14], [15]])

predicted_sales = model.predict(future_months)

print("Existing Sales:", y)

print("Future Months:", future_months.flatten())

print("Predicted Future Sales:", np.round(predicted_sales, 2))

new_month = np.array([[16]])

new_prediction = model.predict(new_month)

print("New Month:", new_month[0][0])

print("Predicted Sales:", round(new_prediction[0], 2))
```

Output:

```
Existing Sales: [120 135 150 165 180 195 210 225 240 255 270 285]
Future Months: [13 14 15]
Predicted Future Sales: [300. 315. 330.]
New Month: 16
Predicted Sales: 345.0
```
